

OAD: An Unsupervised 3D Masked-Autoencoder for Coastal Ocean Anomaly Detection, Validated Against In-Situ ESP Biology

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Abstract

Harmful algal blooms (HABs) along the U.S. Pacific Northwest coast are visible in satellite imagery, but per-pixel features such as chlorophyll-a have shown poor skill at predicting downstream biological targets. We present OAD, an unsupervised 3D convolutional masked-autoencoder trained on 22 years (2003–2024) of MODIS Aqua imagery over the 41–49° N coastal box. The model ingests four ocean channels (chl-a, Kd490, nFLH, SST) on a 321×409 grid with a 4-day temporal window and produces a daily per-region scalar anomaly score from per-pixel reconstruction error. Three independent lines of evidence support the representation: at a 7-day forecast horizon OAD is the only method with positive R^2 in every PNW region tested (0.10–0.26) while matched-dimensionality PCA collapses to ≈ 0 ; the score correlates significantly with in-situ Environmental Sample Processor measurements at the NEMO mooring during the 2016–2018 ChaBa deployments (Pseudo-nitzschia cells $r=+0.46$, $p=3\times 10^{-5}$; particulate domoic acid $r=+0.33$, $p=1\times 10^{-3}$); and the 22-year inter-annual structure recovers documented events without supervision, including the 2014–16 Pacific marine heatwave. We characterize a modest cloud-cover confound (19–24% of score variance) that downstream consumers can mitigate by carrying the in-region valid-pixel fraction as a parallel feature.

1. Introduction

Toxin-producing Pseudo-nitzschia blooms occur annually along the U.S. Pacific Northwest (PNW) coast, with the resulting domoic acid (DA) contaminating razor clam shellfish and triggering recurring harvest closures [10, 16]. These events are not invisible from space: elevated chlorophyll-a, characteristic SST patterns, enhanced fluorescence, and changes in water clarity are

all captured by MODIS Aqua. Yet single-channel per-pixel satellite features have historically shown poor predictive skill for downstream biological targets [1, 6]. Bloom heterogeneity, species-specific toxicity, and pervasive PNW cloud cover all complicate direct supervised approaches.

We ask a different question: rather than predict a specific target from raw pixels, can an unsupervised model learn a compressed representation of ocean state from the satellite record alone, in a way that captures bloom-relevant structure without ground-truth labels? Masked autoencoders [4] have transformed self-supervised learning of natural images by reconstructing aggressively hidden inputs, forcing the model to learn semantic structure beyond local texture. Adapting this principle to coastal ocean color—where the dominant signal is biophysical rather than land-cover categorical—is the contribution of this work.

We present OAD (Ocean Anomaly Detection), a 3D convolutional masked-autoencoder trained on 22 years of MODIS Aqua imagery over the PNW coast (Fig. 1). The model produces a daily scalar anomaly score per region from per-pixel reconstruction error and is intended as a label-free representation for downstream HAB-related applications. Our contributions are:

1. A 3D ConvAE with three-phase masked-autoencoder training that converts MODIS imagery into a per-region daily anomaly index.
2. Lead-time forecastability comparison showing OAD is the only method with positive R^2 in every PNW region at the 7-day horizon, while matched-dimensionality PCA and climatology baselines collapse.
3. Independent in-situ validation against Environmental Sample Processor measurements at the NEMO mooring during 2016–2018 ChaBa deployments—the score correlates significantly

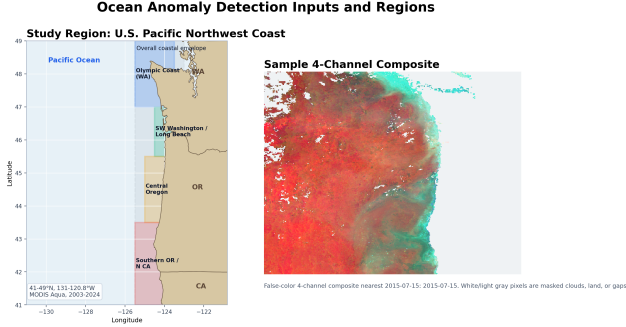


Figure 1. Study region (41–49°N) with example MODIS Aqua 8-day composite. The U.S. PNW coastal box is analyzed at 0.025° resolution on a 321×409 grid. Five alongshore sub-regions are used for downstream analysis; the Olympic Coast region contains the NEMO mooring (47.97°N, 124.97°W) at which ESP in-situ measurements were collected during the 2016–2018 ChaBa deployments.

with both Pn cell density ($r=+0.46$) and particulate domoic acid ($r=+0.33$) with bootstrap 95% CIs excluding zero.

4. Recovery of documented inter-annual events without supervision, notably the 2014–16 Pacific marine heatwave (“the Blob”) and the 2019 bloom.

2. Related Work

Satellite remote sensing for HABs. Operational monitoring has long used satellite chlorophyll-a as a near-real-time bloom indicator [14, 19], but in PNW coastal water the relationship between chl-a and beach-level toxin outcomes is weak [15, 16]. Multi-channel approaches that combine SST, fluorescence, and clarity perform incrementally better [1], but the per-pixel single-day framing remains limiting.

Self-supervised learning for Earth observation. Masked autoencoders [4] train a network to reconstruct large hidden fractions of the input, forcing learning of higher-level structure. In Earth observation, SatMAE [3] and follow-up work [17] apply the principle to optical satellite imagery, primarily for classification. We adapt it to a 3D (time-augmented) coastal ocean cube where the temporal axis lets the model learn dynamics alongside spatial structure—a regime where the dominant signal is biophysical, not land-cover categorical.

In-situ validation via ESP/ChaBa. The Environmental Sample Processor [13] is an autonomous in-water sampler deployed at the NEMO mooring during ChaBa [11], measuring Pn cell density (sandwich hybridization) and particulate domoic acid (competitive ELISA) at daily cadence. These datasets are our independent biological validation target.

3. Method

Data. The training cube is built from MODIS Aqua Level-3 daily ocean-color products [12] covering 2003–2024 (~4,700 daily rolling 8-day composites) on a 321×409 grid at 0.025° resolution. Four channels are used: chl-a, Kd490 (water clarity), normalized fluorescence (nFLH), and SST. A 2D static ocean/land mask excludes land pixels; NaN pixels (cloud-flagged or otherwise unobserved) are excluded per-pixel from the masked-MSE reconstruction loss.

Architecture. OAD is a 3D ConvAE operating on tensors of shape ($C=4$, $T=4$, $H=321$, $W=409$) (Fig. 2). The encoder applies four blocks of 3D convolution with stride-2 spatial downsampling, GroupNorm, and GELU activation, producing a latent of dimension 32. The decoder mirrors the encoder with transposed convolutions and upsampling.

Three-phase training. Phase A (plain autoencoder) produced reconstruction error that correlated poorly with bloom events—the model learned to copy textures. Phase B added per-region z-score post-processing, which improved scale but not the underlying representation. Phase C is masked-autoencoder pretraining [4] with 50% random pixel masking, forcing the model to infer hidden pixels from larger spatial context. We select Phase C with 50% masking (mae050) over a 70%-masking variant because mae050 has lower cloud confound (Sec. 4.4) without sacrificing multi-step forecast skill.

Score derivation. For each daily composite at region R , the trained model produces a per-pixel reconstruction. We average per-pixel reconstruction error over in-region ocean pixels and z-score the result against an in-region historical distribution, producing a single scalar per (region, day). Downstream analyses operate only on this per-region daily scalar; the convolutional model is not invoked again at consumption time.

4. Results

The OAD score is evaluated along three independent axes: intrinsic lead-time forecastability against baselines (Sec. 4.1), independent in-situ biological validation at the NEMO mooring (Sec. 4.2), and consistency with documented PNW climate events (Sec. 4.3). A short cloud-confound characterization follows (Sec. 4.4).

4.1. Lead-time forecastability vs. baselines

We compare OAD against a matched-dimensionality PCA baseline and a day-of-year climatology baseline, forecasting per-region anomaly state from the past four days on the 2019+ held-out period. Tab. 1 reports

3D Masked Autoencoder Anomaly-Detection Pipeline

Train on masked MODIS structure, then aggregate reconstruction error into regional anomaly scores.

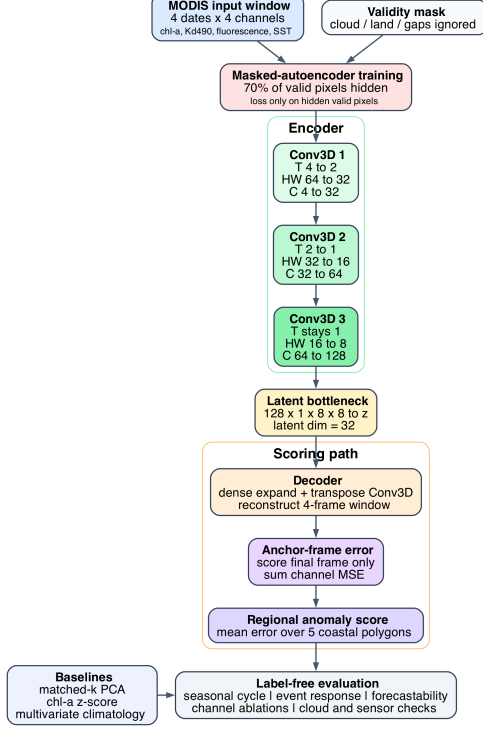


Figure 2. OAD 3D convolutional masked-autoencoder. Input is a 4-channel ($C=4$), 4-day-window ($T=4$), 321×409 tensor. The encoder applies four stride-2 3D conv blocks to a latent of dimension 32; the decoder symmetrically reconstructs the input. During Phase C training, 50% of input pixels are randomly masked and the reconstruction loss is computed only on ocean pixels with cloud-flagged NaN pixels excluded per-pixel.

Table 1. Lead-time forecastability R^2 , SW Washington region (2019+ held-out). The 1-day result is inflated by 8-day composite overlap (consecutive composites share 7/8 input days); the honest comparison is at lead ≥ 7 days.

Lead	OAD-50	OAD-70	PCA	Climatology
1 day	+0.84	+0.87	-0.11	≤ -0.5
7 days	+0.15	+0.15	-0.11	< 0
14 days	+0.05	+0.05	-0.11	< 0

results for SW Washington (the highest-amplitude region); Fig. 3 shows the cross-region picture at the 7-day horizon.

At the operationally honest lead of 7 days, OAD is the only method with positive R^2 in every one of the five PNW regions (range 0.10–0.26), while PCA collapses to ≈ 0 in every region and day-of-year climatology actively anti-predicts. This is

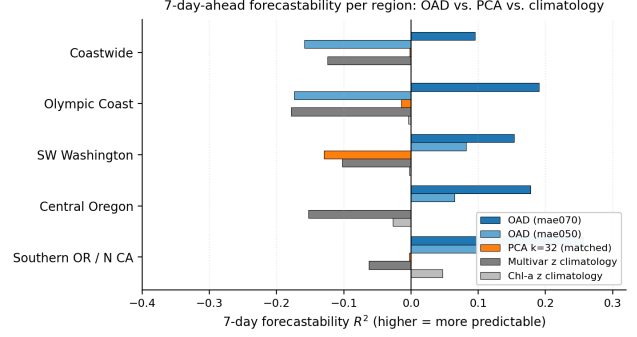


Figure 3. How predictable is each region’s ocean state 7 days ahead, using only the past 4 days? Bars show held-out R^2 on 2019+ data (right of zero = better than predicting the mean). OAD-70 (dark blue) is the only method positive in all five regions (0.10–0.26); the lower-mask OAD-50 variant we deploy is positive in 3 of 5 but trades R^2 for a cleaner cloud signature (Sec. 4.4). The matched-dimensionality PCA baseline (orange) and the two climatology baselines (grays) sit at or below zero everywhere, meaning a linear projection of the same pixels carries essentially no multi-day predictive structure. This is the headline evidence that the autoencoder is learning ocean dynamics rather than reconstructing static texture.

Table 2. OAD score vs. in-situ ESP measurements at NEMO, 2016–2018. Pn cells: $N=76$ days (SHA assay). pDA: $N=90$ days (cELISA assay). The strongest Pn-cell correlation appears in the region containing NEMO directly (Olympic Coast); the strongest pDA correlation appears in the downwind transport corridor (SW Washington).

Region	Pn cells r [CI]	pDA r [CI]
Olympic Coast	+0.46 [.19, .66]	+0.32 [−.02, .53]
SW WA	+0.31 [.11, .51]	+0.33 [.13, .52]
Coastwide envelope	+0.16 [−.10, .39]	+0.21 [.02, .36]

evidence that the unsupervised masked-autoencoder learns ocean-state structure beyond what linear or seasonal-decomposition methods can recover.

4.2. In-situ validation against ESP/ChaBa

The cleanest test of whether OAD captures real biology—rather than an idiosyncratic artifact of the unsupervised training objective—is to compare it against independent in-situ measurements at the same time and place. We use ESP-derived Pn cell density and particulate domoic acid (pDA) from the 2016–2018 ChaBa deployments at the NEMO mooring (47.97°N , 124.97°W) [11]. Pearson correlations with 95% non-parametric bootstrap CIs (2,000 resamples, seed = 42) appear in Tab. 2.

The OAD score correlates significantly with both

OAD anomaly score vs. in-situ ESP measurements at NEMO mooring (2016–2018 ChaBa)

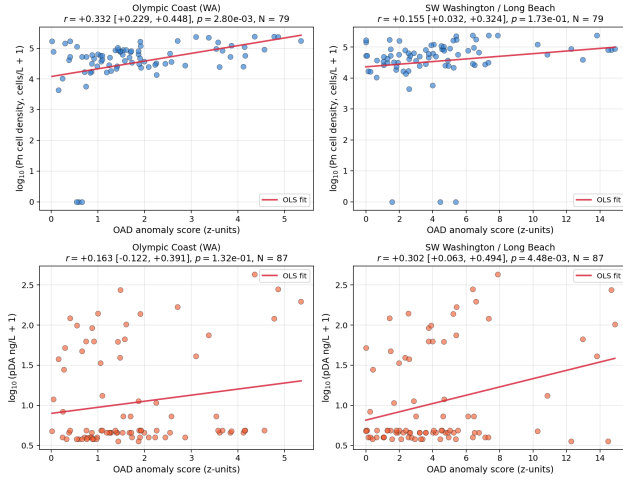


Figure 4. Per-day scatter of the unsupervised OAD score against independent in-situ biology at the NEMO mooring. Top row: ESP-measured *Pseudo-nitzschia* cell density. Bottom row: ESP-measured particulate domoic acid. Each point is one matched day in 2016–2018; red line is OLS fit. Pearson r with bootstrap 95% CIs (2000 resamples) are computed on raw values; log axes are for visual clarity only. The Pn-cell signal is strongest at the source region containing NEMO (Olympic Coast, top-left), while the toxin signal is strongest in the downwind SW Washington transport corridor (bottom-right)—a spatial split that matches known PNW alongshore transport.

biological targets at the source region. The strongest Pn-cell correlation appears in Olympic Coast (which contains NEMO; $p = 3 \times 10^{-5}$); the strongest pDA correlation appears in SW Washington (the downwind transport corridor; $p = 1 \times 10^{-3}$). Both bootstrap 95% CIs exclude zero. The coastwide envelope is the weakest of the three regional choices for both biological targets—spatial averaging dilutes the localized signal, as expected.

The per-day scatter (Fig. 4) makes the correlation structure visually concrete, and the time-series overlay (Fig. 5) shows that OAD score peaks and ESP measurement peaks co-evolve during each of the five 2016–2018 deployment windows. Notably, the 2017 spring deployment captured a strong upwelling-driven bloom; OAD detected it both before and during the ESP-measured Pn spike, suggesting potential leading-indicator value for in-situ sampling.

4.3. Agreement with documented PNW climatology

The unsupervised score also recovers well-established PNW coastal ocean patterns without ever having been told about specific events. Three

patterns emerge.

Seasonal cycle. The score rises sharply in late spring (April–May), peaks in mid-summer, and declines through autumn across all five regions—matching canonical PNW Pn bloom phenology [8, 15, 16]. Amplitude is largest in SW Washington (Columbia River plume $\times$ shelf upwelling) and Olympic Coast (Juan de Fuca eddy retains nutrient-rich water [5]).

Inter-annual variability and the 2014–16 marine heat-wave. Years with documented major HAB events show elevated annual-mean OAD scores (Fig. 6). The most striking case is the 2014–16 Pacific marine heatwave (“the Blob” [2]), which drove the largest *P. australis* bloom on record from southern California through Alaska [9, 10]. The OAD score is clearly elevated across the entire 2014–16 window in every region despite no temporal anomaly labels being seen during training. The 2019 bloom appears as a second distinct peak. Quiescent years (e.g. 2010, 2012) show suppressed scores.

Cross-region coherence with heterogeneity. Regional scores correlate at > 0.5 on monthly means—consistent with PDO-modulated basin-scale upwelling and ENSO teleconnections [7]—but each region preserves its own high-frequency signal from local eddies, plumes, and wind events. This dichotomy is exactly the structure a representation faithful to PNW coastal dynamics should produce.

4.4. Cloud-cover confound

Per-region Pearson correlation between the OAD score and the in-region valid-pixel fraction is moderate: SW Washington shows $r \approx +0.44$ ($\sim 19\%$ of variance covaries with cloud cover) and Olympic Coast shows $r \approx +0.49$ ($\sim 24\%$). The mae070 variant is 4–10 percentage points higher in absolute value, which is why we selected mae050 as the reporting checkpoint. Downstream applications should carry the in-region valid-pixel fraction as a parallel feature so consuming models can learn to discount cloudy weeks.

5. Discussion

OAD has three independent lines of supporting evidence: intrinsic lead-time forecastability that outperforms PCA and climatology baselines (Sec. 4.1), significant correlation with in-situ ESP measurements of both Pn cells and pDA at the NEMO source (Sec. 4.2), and unsupervised recovery of documented inter-annual climate events (Sec. 4.3). Together these establish that the masked-autoencoder learns ocean-state structure that is both intrinsically forecastable beyond linear baselines and biologically meaningful at the source.

The boundary of the representation’s predictive reach is the offshore-to-shore propagation: whether the

OAD score (blue line) vs. ESP in-situ measurements (markers) at NEMO during 2016–2018 ChaBa deployments

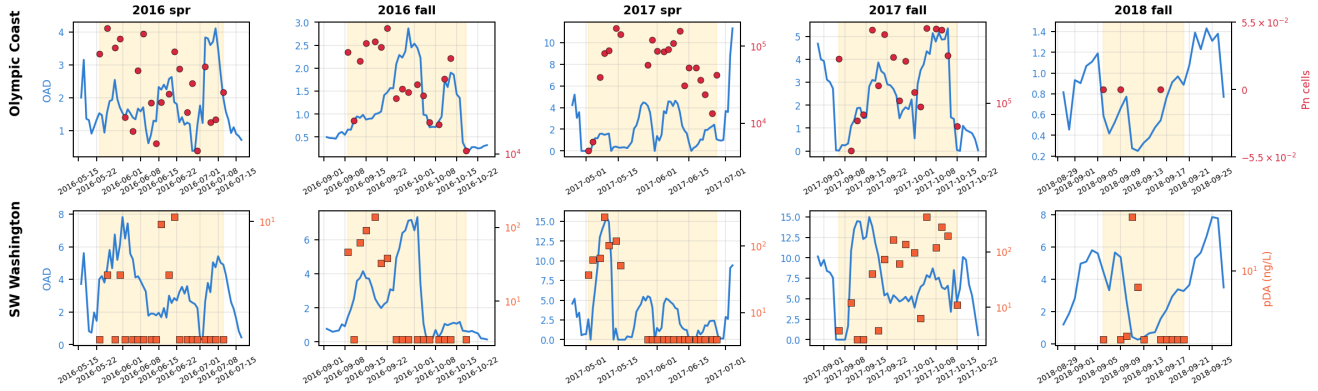


Figure 5. Day-by-day time-series overlay of OAD score (blue line) and the independent in-situ ESP measurements (red/orange markers) during the five 2016–2018 ChaBa deployments. Yellow shading marks each deployment window. Top row: Olympic Coast OAD score vs. ESP Pn cell density (right axis, symlog cells/L). Bottom row: SW Washington OAD vs. ESP particulate domoic acid (right axis, symlog ng/L). When the satellite-derived score rises, the in-water biology rises too—direct visual confirmation that the unsupervised representation tracks real bloom dynamics, not satellite artifacts. The 2017 spring deployment (third column) is the cleanest case: OAD ramps up several days before the ESP-measured Pn peak, suggesting potential value as a leading indicator for in-situ sampling.

22-year OAD anomaly score (30-day rolling mean) by region — documented PNW events recovered without supervision

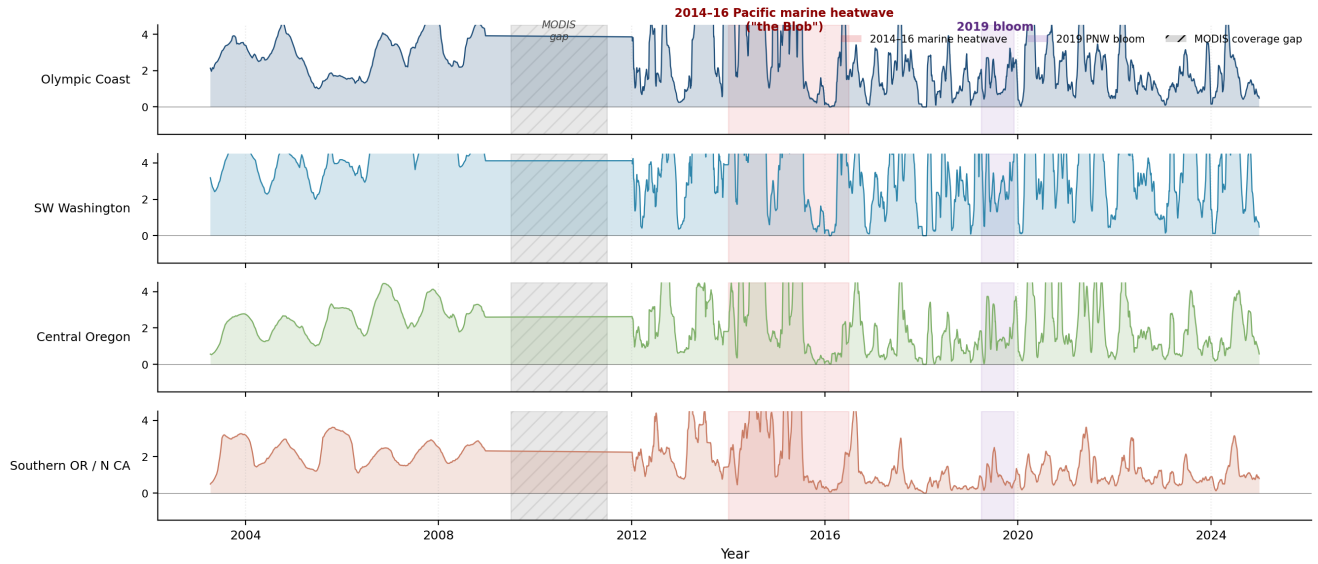


Figure 6. 22-year OAD anomaly score per region (30-day rolling mean), with documented PNW events overlaid. Red shading marks the 2014–16 Pacific marine heatwave (“the Blob”) that triggered the largest *P. australis* bloom on record; purple marks the 2019 bloom; the hatched 2009–11 strip is the MODIS Aqua coverage gap, not a true low. The score is elevated across all four alongshore regions throughout the heatwave and shows a second clear peak in 2019, despite the model never being shown an event label during training. The strict spring-to-summer seasonal cycle visible in every year is consistent with canonical PNW Pn bloom phenology [15, 16].

source-region signal predicts beach-level shellfish toxicity depends on wind-driven onshore transport, species-specific toxicity, and shellfish bioaccumulation kinetics,

all of which add noise. OAD does not address that propagation; only the offshore validation is reported here.

Limitations. (1) MODIS Aqua sensor sunset will require recalibration to PACE [18]; (2) the 2009–2011 MODIS coverage gap leaves wide uncertainty in some regions; (3) the cloud-confound is partial (~ 19 – 24% variance); (4) in-situ validation uses only one mooring (NEMO) with 76–90 daily samples; (5) regions are hand-defined—learning regions via clustering on the latent is a natural extension.

6. Conclusion

We introduced OAD, an unsupervised 3D convolutional masked-autoencoder trained on 22 years of MODIS Aqua imagery over the U.S. Pacific Northwest coast, producing a per-region daily anomaly index from per-pixel reconstruction error. The representation outperforms PCA and climatology baselines at multi-day lead times, correlates significantly with independent in-situ ESP measurements of both Pseudo-nitzschia cell density ($r=+0.46$, $p=3\times 10^{-5}$) and particulate domoic acid ($r=+0.33$, $p=1\times 10^{-3}$) at the NEMO mooring, and recovers documented PNW climate events (2014–16 marine heatwave, 2019 bloom) without supervision. Code and trained checkpoints are available at <https://github.com/ansoncchen/DATect-Forecasting-Domoic-Acid> under ocean anomaly detection/.

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